

Model of
a virusProtective
protein coat

Viruses

Viruses are much, much smaller than bacteria. They are shaped like spheres or rods. Viruses are not really alive, because they are not made of cells. They only become active when they invade a cell. They copy themselves by taking over the cell and turning it into a virus factory.

DNA or
RNA strand

Plant viruses

Plant viruses can change the way that plants develop. For example, one virus affects the pigment in tulips' petals. It stops the pigment

from working in some places. This makes the petals look striped.



A virus has made light patches appear on these leaves.



The streaked patterns on this tulip are caused by a virus.

Vaccinations

Vaccinations can help to protect people from harmful diseases. A person is injected with a weakened form of a virus or bacterium. This prepares the immune system for the real thing.



Harmful viruses

Viruses can cause different illnesses.



Chickenpox is easy to catch. The main symptom is itchy sores.



Rabies is a fatal virus that is common in animals such as dogs.



Colds are viruses and can bring on a sore throat, runny nose, and cough.



Microscope photograph of algae

Other tiny cells

Like bacteria, another group of organisms called archaea are all single-celled. There are also some single-celled organisms, such as amoebas, algae, and yeasts, that are closely related to animals, fungi, and plants.